

MUHAMMAD HARIS

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AI/ML Engineer | Generative AI | ML Systems | Robotics & Research

PROFESSIONAL SUMMARY

AI Research Engineer and Robotics & AI master's student at the (NUST), specializing in machine learning, generative AI, and large language model systems. Experienced in designing reproducible machine learning experiments, evaluating deep learning models, developing AI inference pipelines, and building modular AI systems from data ingestion through deployment.

Hands-on experience with PyTorch, TensorFlow, Hugging Face, LangChain, LlamaIndex, and modern retrieval-augmented generation (RAG) systems. Passionate about model evaluation, benchmark design, generative modelling, and developing reliable AI systems that bridge research and production.

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow/Keras, scikit-learn, HuggingFace, KerasNLP, LangChain, LlamaIndex, Stable Diffusion, YOLOv5, VAE, RAG pipelines, LLM integration, feature engineering, model evaluation, inference optimization

Programming / Systems: Python, C++, Node.js, Docker, Linux, REST APIs, SQL, Git/GitHub, Google Colab, Flutter/Dart

Robotics / Research: ROS, MATLAB/Simulink, Arduino, computer vision, generative modeling, simulation, experiment design, self-play

EXPERIENCE

AI Software Developer / *DevCrafter AI* | 2025 - Present

- Designed an end-to-end ML inference pipeline for learning-style classification, ingesting quiz responses and behavioral activity features, processing them through a trained model, and returning personalized profiles through a structured REST API.
- Built reusable preprocessing and inference flow for comparative analytics, keeping the service logic modular enough for future model swaps and production hardening.
- Developed maintainable AI service architecture supporting future model evaluation and deployment workflows.

RESEARCH & THESIS

Thesis Research: Generative AI for Biomimetic Prosthetic Hand Grasping / *NUST* | 2024 - Present

Research focuses on generative modelling for prosthetic hand control using conditional VAEs and diffusion models.

- Designing reproducible ML experiments
- Building structured datasets for grasp generation
- Evaluating generative model quality using simulation
- Conducting ablation studies across architectures
- Analyzing failure cases and model behavior
- Comparing candidate models using quantitative evaluation metrics

SELECTED AI / ML PROJECTS

CortexOS | AI Knowledge System

- Building a modular AI knowledge system that helps large language models search, organize, and answer questions from documents instead of relying only on pretrained memory.
- Designed reusable ingestion, retrieval, and indexing pipelines using LangChain, LlamaIndex, embeddings, and clean software architecture.
- Focused on making the system extensible for research assistants, document Q&A, and enterprise knowledge workflows.

Biomimicking Gait Analysis / *Final Year Project*

- Built a computer vision pipeline with YOLOv5 to analyze amputee gait and joint angles during walking.
- Compared extracted motion patterns with normal gait to identify deviations and generated reports and visualizations for rehabilitation and prosthetic design support.

Predicting Heart Failure Using Machine Learning Models

- Built and compared Logistic Regression, SVM, Decision Tree, KNN, and Random Forest models for clinical risk prediction.
- Applied preprocessing, cross-validation, and feature analysis to reach 90.5% accuracy and surface clinically useful features.

Stable Diffusion-Based Text-to-Image Generation

- Built a GPU-accelerated text-to-image inference pipeline using Stable Diffusion v1.5, PyTorch, and HuggingFace.
- Used prompt tuning and attention slicing to improve generation efficiency and stability.

Variational Autoencoder for Structured Latent-Space Learning

- Implemented a custom VAE on Fashion-MNIST with encoder-decoder networks, KL regularisation, and the reparameterisation trick.
- Visualised latent space structure and generated synthetic samples to study unsupervised representation learning.

Natural Language Processing with Disaster Tweets

- Worked on the Kaggle disaster tweet classification task using Keras and KerasNLP.
- Experimented with RNNs and transformer-based architectures for noisy social text classification.

Checkers Game with AI

- Developed an 8x8 checkers engine using Minimax and a custom evaluation function.
- Added self-play learning and a Pygame interface for experimentation and visualisation.

ADDITIONAL SYSTEMS ENGINEERING

C++ Financial Risk & Derivatives Pricing Engine | Pricing, market-data infrastructure, and reusable architecture

- Built a high-performance C++ engine supporting Black-Scholes, CRR binomial tree pricing for European and American options, bond valuation, and swap pricing.
- Implemented a decoupled market-data ingestion layer for yield curves, volatility surfaces, and spot interpolation so pricing logic stayed reusable and testable.

EDUCATION

MS Robotics & AI | NUST | 2024 - Present